

# **Quality of Discretionary Environmental Disclosure, Corporate Governance and Market Value Creation: a Panel Data Analysis in Indian Context**

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# **Quality of Discretionary Environmental Disclosure, Corporate Governance and Market Value Creation: a Panel Data Analysis in Indian Context**

## **Abstract**

The main focus of this study is to examine how corporate governance practices of companies influence voluntary environmental disclosure quality of firms from polluting as well as non-polluting sectors in Indian context. The study further investigates how voluntary environmental disclosures influence long term as well as short term market value creation of the firm. The study reveals significant impact of corporate governance variables on corporate voluntary environmental disclosures. On the other hand our model reveals that market discounts environmental disclosure information in the short term value creation but no significant association is found between voluntary environmental disclosure and sustained market value creation of firms in polluting sectors. Information technology and enabled services as a non-polluting sector reports negative association in regard to voluntary environmental disclosure and long term market value creation.

## **Introduction**

The disclosure of corporate environmental practices largely remains unregulated (Kolk, 2008) and voluntary. The managerial decisions about the extent of voluntary disclosure to be made by the firm are influenced by board of directors and shareholders (Millstein, 1991). This phenomenon of voluntary disclosure of corporate environmental information in annual reports and sustainability or environmental reports has resulted in substantial literature in this field of study (Campbell, 2004). Tietenberg (1998) has noted three waves on environmental practices; first wave dealt with legal remedies in the form of prohibition or emission standards, second wave was market oriented, like charges for emission, permits, subsidies The third or the most recent wave of environmental regulation has been the voluntary disclosure of corporate environmental information to the public. KPMG (2008) in its 'International survey on corporate

sustainability reporting' outlines that in order to meet the stakeholders' demand for accountability and transparency more and more firms are increasingly relying on voluntary disclosure of environmental practices.

There has been extensive research on the determinants of corporate environmental disclosures in European and North American Perspectives (Aerts, Cormier, & Magnan, 2008), but the research on voluntary environmental disclosures in developing countries, though increasing, requires more intensive investigation (Elijido-Ten, 2011). In this study we present the Indian context for voluntary environmental disclosure among polluting and non-polluting industries. To specify our model in the developing country context we have extensively drawn from earlier literature mostly from developed as well as developing countries.

Earlier corporate environmental disclosures comprised some pollution abatement related documents. Insignificant voluntary disclosures were part of the financial statements until it emerged as a communicative action that improves corporate accountability leading to stakeholders' empowerment to take part into discourse, dialogue and discussion (Cooper & Owen, Corporate social reporting and stakeholder accountability: the missing link, 2007). The theoretical grounding of disclosure literature is primarily explained with stakeholder theory, legitimacy theory and political economy theory (Huang & Kung, 2010; Bewley & Li, 2000; Deegan & Gordon, A study of the environmental disclosure practices of Australian corporations, 1996). The initial phase of empirical study on voluntary environmental disclosures started in 1970s when the main question was whether it economically pays to be socially and environmentally responsible and to find that, researchers have tried to associate voluntary disclosure with firm characteristics (King & Lenox, 2001; Konar & Cohen, 2001). A broad category of studies have isolated the market based measures of firm performances and studied

the causality between voluntary environmental disclosures and accounting based measures of firm performances (Margolis & Walsh, 2003). The results are inconclusive. A few studies on the causality between voluntary environmental disclosures and accounting based measures of firm performances reported positive association (Clarkson, 1988; Hart & Ahuja, 1996; Judge & Douglas, 1998; Dowell, Hart, & Young, 2000; Graves & Waddock, 2000), while a few others reported mixed results (Graves & Waddock, 1994; Rockness, Schlachtr, & Rockness, 1986) as well as no significant association (Ingram & Frazier, 1983). In comparison to this, the causality studies with market based measures are more consistent. Using US sample Belkaoui (1976) reported positive association between pollution control information and stock price which is in line with the findings of Shane and Spicer (1983) and Cormier et al. (1993). On risk management perspective higher environmental performance and disclosure led to the reduced volatility with stock market returns (Blacconiere & Patten, 1994; Hamilton, 1995). The positive association between corporate environmental performance, disclosure and stock market responsiveness is further established in US market (Al-Tuwaijri, Christensen, & Hughes II, 2004; Cohen, Fenn, & Konar, 1997) and Greece Market (Halkos & Sepetis, 2007). On the other hand the contradictory outcomes include the study by Lanoie et al (1998) showing no relationship and Hassel et al. (2005) reporting negative association between environmental disclosure and stock market returns in Stockholm stock exchange.

Looking at the inconclusiveness of the earlier studies two things seems to be responsible. First, as the disclosure is voluntary, so there exists substantial difference in disclosure practices among companies (Walden & Schwartz, 1997; Patten, Exposure, legitimacy and social disclosure, 1991; Patten, Intra industry environmental disclosures in response to Alaskan oil

spills: a note on legitimacy theory, 1992). Secondly, Horvathova (2010) in a meta-analysis pointed out towards some omitted variables for these varying outcomes.

In earlier literature limited attention has been paid on the aspects of corporate governance as the determinants of voluntary disclosures of corporate environmental practices. Moreover the study on the impact of corporate governance on voluntary environmental disclosures in the developing country context has been scant. Looking at the inconclusiveness of earlier studies on voluntary environmental disclosure and accounting based measures of firm performances we have taken firm characteristics as the control variables and corporate governance variables are taken as explanatory variables. Further to test its association with market based measures we have taken Tobin's Q as well as price to earnings ratio and tried to estimate how voluntary environmental disclosures influence these two market based measures. These two variables are quite different as the quotient in Tobin's Q measures long term value creation by the firm whereas P/E multiple is a short term measure of value creation. We hope this estimation will throw some light on the context of getting varying outcome in earlier studies. Based on the above, in the following section we present our research hypothesis.

### **Corporate Governance and Voluntary Environmental Disclosures:**

Corporate governance includes various mechanisms to increase transparency and protect stakeholders' interest mainly by exercising stakeholders control over management (Ingley & Van Der Walt, 2004). The theoretical basis of the study on the impact of corporate governance on the voluntary environmental disclosures is primarily based on agency theory (Jensen & Meckling, 1976) but stakeholder theory and legitimacy theory also provides some reliable framework (Huse, 2005) that explains disclosures as the communicative action to gain organizational legitimacy by engaging stakeholders (Unerman & Bennett, 2004). However this approach is

criticized as a strategy for stakeholders' management instead of stakeholders' engagement (Cooper & Owen, 2007).

The attempts to relate corporate governance parameters as determinants of corporate environmental performances have been limited (Michelon & Parbonetti, 2010; Williamson, 1984). Further, earlier studies mostly have been done in American and European context (Forker, 1992; Cerbioni & Parbonetti, 2007) and comparatively limited in Asian Context, like Leung and Horwitz (2004) in Hong kong, Cheng and Courtney (2006) in Singapore, Latridis (2013) in Malaysia among a few prominent studies.

The first aspect of this paper is to measure the quality of corporate environmental disclosure practices in polluting as well as non-polluting sectors in Indian context and also to understand how voluntary environmental disclosure is influenced by corporate governance practices that are based on the principles of corporate accountability, probity and transparency. Following are the testable hypotheses based on corporate governance variables.

#### Board Independence:

It is well accepted that higher proportion of independent directors can monitor the management more effectively (Liao, Luo, & Tang, 2014) to make sure that they act not only on behalf of the shareholders but also consider the interest of other stakeholders. Internal and independent directors have different incentives and values (Donnelly & Mulcahy, 2008) and moreover independent directors have stronger stakeholder orientation due to their lack of financial stake in the firm and diverse background (Wang & Dewhirst, 1992).

***Hypothesis 1: Voluntary environmental disclosure is positively related with the board independence.***

### Ownership Concentration:

In the corporate governance literature ownership concentration has been used in limited number of studies. For some studies it has been describes as family control or managerial ownership where owners are the executive management of the company (Chau & Gray, Family ownership, board independence and voluntary disclosure: evidence from Hong Kong, 2010; Khan, Muttakin, & Siddiqui, 2013). In our sample of companies with large market capitalization ‘CEO duality’ can play this role but we are interested to see what happens to the environmental disclosure for largely concentrated ownership.

***Hypothesis 2: Higher ownership concentration is negatively related with voluntary environmental disclosure.***

### CEO Duality:

Sometimes promoters take the dual role of board chairman as well as the executive head (CEO/MD) of the company. A role duality may lead to hiding of information from the non-executive independent directors of the board (Haniffa & Cooke, Culture, corporate governance and disclosure in Malaysian corporations, 2002).

***Hypothesis 3: CEO duality is negatively related to voluntary environmental disclosures***

### Public Shareholding:

Ownership distributioin among many investors increases the pressure for voluntary disclosure of the firm (Chau, & Gray, 2002). General public are assumed to be more sensitive to the media exposure and activism of the public interest groups or NGOs.

***Hypothesis 4:*** *Larger proportion of shareholding among general public leads to higher level of voluntary environmental disclosures.*

Environmental Committee:

A very limited number of companies in India have a formal environmental committee in the board. An environmental committee may increase the environmental awareness among employees and their impact of the job (Liao, Luo, & Tang, 2014).

***Hypothesis 5:*** *The propensity of voluntary environmental disclosure is positively associated the presence of environmental committee.*

#### **Voluntary Environmental Disclosure and Market Value Creation:**

The varying results of the studies on the environmental disclosures and market value creation may be due to the difference in the variables taken to satisfy different objectives of the studies. For example, release of environmental news is a short term phenomena and will impact the market accordingly whereas forming an environmental committee in the board has long term implications and market will discount the information not in the same manner like the release of environmental news. In this study we have taken Tobin's Q as a measure of long term value creation by the firm. Latridis (2013) studied common-law Malaysia and reported positive association between corporate environmental disclosure and Tobin's Q. On the other hand price to earnings ratio is considered as a measure of the short term multiple of value creation. In this study we try to estimate how long term as well as short term value creation measures react to voluntary environmental disclosures. Studying the literature above we propose the hypothesis in the following way.



***Hypothesis 6:*** *Higher voluntary environmental disclosures lead to higher levels of Tobin's Q both in polluting as well as non-polluting sectors.*

***Hypothesis 7:*** *Voluntary environmental disclosures are positively associated with P/E multiple both in polluting as well as non-polluting sectors.*

**Dataset, Environmental Disclosure Index and Model Specification:**

**Dataset:**

Our study comprises sample from polluting as well as non-polluting sectors in Indian context. Referring Ministry of Environment and Forests, Govt. of India we have identified cement and Iron & steel industry as these are categorized as heavily polluting and covered under 'Red' category of the Central Action Plan. On the other hand information technology and enabled services (IT&ES) is taken as a non-polluting sector. Our primary sampling design was to take to thirty market capitalization Company from each sector for the time period of 2008-09 to 2012-13, that is five years of study period. But due to inconsistency and incomparability of data for some of the companies we had reduce our sample and the final sample comprised of 26 companies from cement industry, 29 companies from iron and steel industry and 29 companies from IT & ES industry. So the total sample comprised of 84 companies distributed in cement, iron and steel and information technology and enabled services sector for the study period of five years.

**Environmental Disclosure Index:**

The key issue of the research design in the present study is to construct a reliable proxy for the quality of voluntary environmental disclosure of the firm (Roy & Ghosh, 2011). Environmental disclosures on print based reports that is, annual report, sustainability report and environmental report are considered for the purpose of the study. Content analysis methodology

is used to measure the extent of disclosure. For construction of the disclosure grid we have extensively referred Global Reporting Initiative's G4 guideline on discretionary disclosure of corporate environmental practices as it is the most widely accepted voluntary disclosure guidelines for large companies (KPMG, 2008). For scoring and index construction methodology of voluntary environmental disclosures using content analysis we have further referred Aerts, Cormier and Magnan (2007) and Al-Tuwaijri et al (2004). The detail scoring method is given in the Annexure of this paper.

#### Model Specification:

At first we want to test the impact of corporate governance on the voluntary environmental disclosures controlling for firm characteristics. The following model structurally represents the relationship that is adopted for the empirical analysis.

#### Model 1:

$$(\text{Environmental Disclosure})_{it} = f(\text{Corporate Governance, Firm Characteristics})_{it}$$

The study uses a panel data methodology to test the model. Following Hausman test, the estimation of the panel data using fixed effect model is found more consistent than GLS. So, we have used the Fixed Effects Panel Data analysis method using an unbalanced panel data set of 26 sample companies from cement industry, 29 companies from iron and steel industry and 29 companies from IT&ES industry over 5 years study period. We consider firm specific stakeholders' sensitivity as the unobserved firm effect ( $\alpha_i$ ) which is fixed in the study period and

correlated with the explanatory variables. The working model with unobserved effects is as follows:

$$\begin{aligned} (\text{Environmental Disclosure})_{it} = & \alpha_0 + \alpha_1 D_1 + \alpha_2 D_2 + \alpha_3 D_3 + \alpha_4 D_4 + \beta_1 (\text{Independent directors})_{it} + \\ & \beta_2 (\text{Ownership concentration})_{it} + \beta_3 (\text{CEO Duality})_{it} + \beta_4 (\text{Public Shareholding})_{it} \\ & + \beta_5 (\text{Environmental Committee})_{it} + \beta_6 (\text{Leverage})_{it} + \beta_7 (\text{Capital intensity})_{it} + \\ & \beta_8 (\text{Age of fixed assets})_{it} + \beta_9 (\text{Firm size})_{it} + \beta_{10} (\text{ROA})_{it} + a_i + u_{it} \end{aligned}$$

The second aspect of this paper is to study the impact of voluntary environmental disclosures on market value creation, both short term as well as long term in nature, controlling for firm characteristics. So the functional relationship is presented as follows:

Model 2

$$(\text{Market Value of Firm})_{it} = f(\text{Environmental Disclosure, Firm Characteristics})_{it}$$

For long term and short term measure of market value creation we have taken Tobin's Q and P/E multiple respectively. So the working models with longitudinal dataset from the same sample with unobserved fixed effect are as follows.

$$\begin{aligned} (\text{Tobin's } Q)_{it} = & \alpha_0 + \alpha_1 D_1 + \alpha_2 D_2 + \alpha_3 D_3 + \alpha_4 D_4 + \beta_1 (\text{Environmental Disclosure})_{it} + \\ & \beta_2 (\text{Leverage})_{it} + \beta_3 (\text{Capital intensity})_{it} + \beta_4 (\text{Age of fixed assets})_{it} + \\ & \beta_5 (\text{Firm size})_{it} + \beta_6 (\text{ROA})_{it} + a_i + u_{it} \end{aligned}$$

For short term measures Tobin's Q is replaced by P/E multiple in the following model.

$$\begin{aligned} (\text{P/E Ratio})_{it} = & \alpha_0 + \alpha_1 D_1 + \alpha_2 D_2 + \alpha_3 D_3 + \alpha_4 D_4 + \beta_1 (\text{Environmental Disclosure})_{it} + \\ & \beta_2 (\text{Leverage})_{it} + \beta_3 (\text{Capital intensity})_{it} + \beta_4 (\text{Age of fixed assets})_{it} + \\ & \beta_5 (\text{Firm size})_{it} + \beta_6 (\text{ROA})_{it} + a_i + u_{it} \end{aligned}$$

The fixed effect panel data model is estimated with time-demeaned or within transformation method. Here  $D_1$  to  $D_4$  are four consecutive time dummies for five year study period taking 2008-09 as the base year. The measurements of the variables are as follows:

Independent directors: proportion of independent non-executive directors in the board

Ownership concentration: percentage shareholding of the promoter group

CEO Duality: It is a dichotomous variable. Takes the value 1 if CEO is also the chairman of the board and 0 otherwise

Public shareholding: Percentage shareholding with public

Tobin's Q: Market value of equity plus book value of preference shares plus book value of long term debt scaled by total asset.

Leverage: long term debt to equity ratio

Capital intensity: percentage of net block to total asset

Age of fixed assets: ratio of accumulated depreciation to gross block

Firm size: Ln (asset)

Return on assets: percentage of PBIT to Net block

### **Results and Analysis:**

The overall mean disclosure score is quite low for Indian companies considered for this study. Out of total possible score of 60, the highest mean disclosure score from iron and steel industry stood at 25.15 followed by IT&ES (21.08) and cement (20.06) industry.

### **Insert Table – 1 here**

Firms scored very low on the parameters of assurance, governance, commitments and engagements. Environmental performance indicators are also rudimentary in nature. But, the

disclosure score reveals that for all the three sectors the average disclosures are gradually increasing over the study period.

**Insert Table – 2 here**

The descriptive statistics further reveals the variability of disclosures among IT & ES companies are highest with standard deviation being 16.688. As per our expectation the average value of Tobin's Q in ITES industry is much higher (17.724) than cement (2.564) and iron and steel (2.811) industry. P/E ratio in the cement industry has outperformed others (24.99) but at the same time it is more sensitive also bearing the highest value of standard deviation (28.24). Among corporate governance variables, ITES companies scored highest in terms of board independence (54.625) which reflects the culture of openness to ideas among technology companies. Ownership concentration in the polluting sectors under sample are quite same but the ownership is slightly less concentrated in ITES companies (50.281). Cement industry is highest in terms of public shareholding (18.10) and ITES scored lowest (15.95) on this parameter reflecting greater influence of institutional investors. The correlation matrixes of the variables of three sectors are given separately in Table 3a, 3b and 3c.

**Insert Table – 3a, 3b, 3c here**

The result of the fixed effect panel data is quite prominent for our first working model. We have shown the result separately for all three sectors under study and the pooled sample of polluting sectors. The first point to be noted here is the coefficient of dummy variables. The values are significant and coefficients are gradually increasing which indicates general increase in the value of disclosure over the study period in compared to the base year of 2008-09. Independent director is showing positive but insignificant association among all the sample companies. This result is as per our predicted value but in contrary to the results of Michelin and

Parbonetti (2010) that shows negative association. Our study indicates that independent directors try to engage companies in sustainability disclosures to ensure corporate legitimacy (Tricker, 1984; Haniffa & Cooke, The impact of culture and governance on corporate social, 2005) but cannot substantiate as the values are not statistically significant. As predicted ownership concentration is negatively associated with voluntary environmental disclosure and the values are significant at 5 % level for ITES and iron and steel industry. Greater public shareholding leads to higher exposure. Cement industry has highest concentration of public shareholding and reported significantly positive association with voluntary environmental disclosure. Presence of the environmental committee has significantly positive impact on disclosure only in case of the cement industry. We expected that firms initially try to satisfy its shareholders and so concentrates on voluntary disclosures only if it has earned its expected profit. So we predicted higher profit will lead to higher voluntary disclosures. Contrary to our expectation the result reports negative associations between ROA and disclosure for our first working model and the result from cement and iron and steel industry are statistically significant. This indicates a simple fact that the resource allocation on corporate environmental practices are a charge against profit and higher environmental practices leads to higher level of voluntary environmental disclosures (Al-Tuwaijri, Christensen, & Hughes II, 2004).

#### **Insert Table – 4 and 5 here**

In market based measures we have found quite interesting results studying the impact of voluntary environmental disclosures on long term as well as short term market based performance measures. Contrary to our prediction, for cement industry environmental disclosure has a negative and statistically insignificant impact on Tobin's Q whereas when it comes to short term measures environmental disclosure shows statistically significant positive results with P/ E

ratio. This shows in a polluting industry like cement, market is sensitive for environmental information disclosure in the short run but the environmental disclosure information has no significant impact on Tobin's Q. Same phenomena is evident in iron and steel industry also. In non-polluting ITES sector the market has considered environmental performance and disclosure as misallocation of resources and reports significant negative association both with Tobin's Q as a measure of sustained value creation and P/E multiple as a measure of short term market impact. In terms of firm characteristics, cement industry has reported significant negative association among age of fixed assets, firm size with Tobin's Q as well as P/E multiple. Further P/E multiple reports significant positive association with ROA in the sample of cement industry. Firm size further reports significant negative association with Tobin's Q in iron and steel industry and with P/E multiple in ITES industry. As per our expectation ROA in ITES sector is positively associated with P/E ratio.

**Insert Table – 6a and 6b here**

### **Concluding Remarks:**

Corporate governance represents the culture and philosophy of the company and thus proved to be a determining factor for voluntary environmental disclosures. The overall voluntary disclosures of corporate environmental performances are quite limited in Indian context. Public pressure is major factor for determining firms' disclosure strategy as the higher public shareholding leads to better disclosure strategy. The study reveals that high ownership concentration and leverage are detrimental to voluntary environmental disclosures. When it comes to market based measures market clearly discounts environmental information as a short term phenomena. Moreover non-polluting industry also gets negatively impacted from

environmental information disclosure in short term and also discourages the practice as a misallocation of fund for sustained market value creation. Further the study could have been stretched to other industries. More insights are expected in case of cross country comparisons.



## Tables

**Table – 1: Environmental disclosures mean score by industry**

|                                                | Cement Industry | Iron and steel Industry |              |              | IT&ES Industry |              |
|------------------------------------------------|-----------------|-------------------------|--------------|--------------|----------------|--------------|
| Samples companies                              | 26              | 29                      |              |              | 29             |              |
| Disclosure Grid                                |                 | 2008 – 09               | 2009 – 10    | 2010 – 11    | 2011 – 12      | 2012 - 13    |
| Strategy and Analysis                          | Cement          | 2.35                    | 2.4          | 2.58         | 2.62           | 2.85         |
|                                                | Iron & steel    | 2.90                    | 3.10         | 3.17         | 3.38           | 3.55         |
|                                                | IT&ES           | 2.74                    | 2.68         | 2.79         | 2.90           | 2.97         |
| Assurance                                      | Cement          | 0.17                    | 0.24         | 0.23         | 0.31           | 0.35         |
|                                                | Iron & steel    | 0.17                    | 0.17         | 0.21         | 0.28           | 0.28         |
|                                                | IT&ES           | 0.30                    | 0.29         | 0.31         | 0.31           | 0.31         |
| Governance, Commitments and Engagements        | Cement          | 0.61                    | 0.72         | 0.77         | 1.00           | 1.00         |
|                                                | Iron & steel    | 1.03                    | 1.14         | 1.31         | 1.62           | 1.72         |
|                                                | IT&ES           | 0.89                    | 0.89         | 1.10         | 1.10           | 1.21         |
| Laws and Regulations                           | Cement          | 0.17                    | 0.2          | 0.23         | 0.23           | 0.62         |
|                                                | Iron & steel    | 0.57                    | 0.62         | 0.79         | 0.97           | 1.17         |
|                                                | IT&ES           | 0.926                   | 0.96         | 1.21         | 1.41           | 1.55         |
| Expenditures and Risks                         | Cement          | 3.30                    | 3.72         | 3.92         | 4.38           | 4.81         |
|                                                | Iron & steel    | 3.47                    | 4.03         | 4.72         | 5.21           | 5.62         |
|                                                | IT&ES           | 3.111                   | 3.36         | 3.79         | 4.14           | 4.34         |
| Environmental Performance Indicators           | Cement          | 9.91                    | 11.32        | 12.35        | 13.23          | 13.69        |
|                                                | Iron & steel    | 12.57                   | 13.97        | 15.21        | 15.90          | 16.93        |
|                                                | IT&ES           | 10.704                  | 10.75        | 12.03        | 12.97          | 13.34        |
| Mean score of disclosure                       | Cement          | <b>16.52</b>            | <b>18.6</b>  | <b>20.08</b> | <b>21.77</b>   | <b>23.31</b> |
|                                                | Iron & steel    | <b>20.70</b>            | <b>23.03</b> | <b>25.41</b> | <b>27.34</b>   | <b>29.28</b> |
|                                                | IT&ES           | <b>18.67</b>            | <b>18.93</b> | <b>21.24</b> | <b>22.83</b>   | <b>23.72</b> |
| Overall mean Score of Environmental Disclosure | Cement          | <b>20.06</b>            |              |              |                |              |
|                                                | Iron & steel    | <b>25.15</b>            |              |              |                |              |
|                                                | IT&ES           | <b>21.08</b>            |              |              |                |              |

**Table - 2: Summary statistics of Variables**

| Variables                       | Cement Industry |         |       |         | Iron and steel |         |        |         | IT&ES Industry |         |          |         |
|---------------------------------|-----------------|---------|-------|---------|----------------|---------|--------|---------|----------------|---------|----------|---------|
|                                 | Mean            | Std Dvn | Min.  | Max.    | Mean           | Std Dvn | Min.   | Max.    | Mean           | Std Dvn | Min.     | Max.    |
| <b>Environmental Disclosure</b> | 20.06           | 12.37   | 7.000 | 54.00   | 25.15          | 13.747  | 7.00   | 60.00   | 21.08          | 16.668  | 3.0000   | 59.000  |
| <b>Tobin's Q</b>                | 2.564           | 2.336   | 0.477 | 13.0353 | 2.81153        | 2.8366  | 0.5784 | 21.2393 | 17.724         | 20.034  | 0.64549  | 93.370  |
| <b>P/E Ratio</b>                | 24.99           | 28.24   | 0.570 | 176.43  | 12.8105        | 16.11   | 0.20   | 104.52  | 14.039         | 9.4499  | 0.200    | 53.330  |
| <b>Independent Director</b>     | 52.61           | 10.86   | 25.00 | 72.73   | 50.37          | 7.97    | 22.0   | 75.0    | 54.625         | 11.689  | 22.222   | 83.333  |
| <b>Ownership Concentration</b>  | 53.44           | 14.20   | 25.49 | 77.21   | 53.65          | 13.05   | 25.27  | 85.82   | 50.281         | 19.768  | 6.8300   | 83.320  |
| <b>Public Shareholding</b>      | 18.10           | 12.92   | 1.08  | 50.11   | 17.05          | 9.99    | 1.84   | 44.62   | 15.954         | 11.475  | 2.5200   | 49.640  |
| <b>Leverage</b>                 | 1.25            | 3.07    | 0.0   | 23.89   | 0.79           | 0.81    | 0.00   | 3.93    | 0.19887        | 0.44124 | 0.010000 | 1.9800  |
| <b>Capital Intensity</b>        | 60.13           | 27.79   | 1.47  | 202.72  | 49.56          | 31.65   | 4.88   | 243.08  | 20.645         | 14.987  | 0.15603  | 88.444  |
| <b>Age of Fixed Assets</b>      | 0.40            | 0.17    | 0.06  | 0.75    | 0.34           | 0.17    | 0.00   | 0.70    | 0.46144        | 0.21391 | 0.00     | 0.88256 |
| <b>Firm size</b>                | 5.88            | 1.76    | 2.41  | 9.48    | 6.90           | 1.58    | 3.68   | 10.23   | 5.0145         | 1.9571  | 0.040822 | 8.5378  |
| <b>ROA</b>                      | 33.54           | 149.6   | -1309 | 787.81  | 31.72          | 25.83   | -13.14 | 139.77  | 219.68         | 574.78  | -12.220  | 4920.8  |

**Note:** CEO Duality and Environmental Committee are dichotomous variables and thus not considered in summary statistics

**Table 3a: Correlation Matrix (Cement Industry)**

|                          | Environmental Disclosure | Tobin's Q | P/E Ratio | Independent director | Ownership Concentration | Public Shareholding | Leverage | Capital Intensity | Age of Fixed Asset | Firm size | ROA      |
|--------------------------|--------------------------|-----------|-----------|----------------------|-------------------------|---------------------|----------|-------------------|--------------------|-----------|----------|
| Environmental Disclosure | 1.0000                   | 0.3598    | -0.0312   | 0.2570               | - 0.1754                | -0.3004             | -0.2147  | -0.0468           | 0.2487             | 0.6161    | 0.0641   |
| Tobin's Q                |                          | 1.0000    | 0.1582    | NA                   | NA                      | NA                  | 0.0713   | -0.5606           | 0.2959             | - 0.0626  | 0.1308   |
| P/E Ratio                |                          |           | 1.0000    | NA                   | NA                      | NA                  | -0.0161  | -0.0959           | -0.0743            | - 0.1380  | 0.0487   |
| Independent director     |                          |           |           | 1.0000               | 0.0802                  | -0.2644             | -0.0156  | -0.1573           | -0.0161            | 0.1780    | 0.1345   |
| Ownership Concentration  |                          |           |           |                      | 1.0000                  | -0.1929             | 0.1130   | 0.1296            | 0.0837             | 0.2008    | 0.0109   |
| Public Shareholding      |                          |           |           |                      |                         | 1.0000              | -0.0954  | -0.1167           | 0.1877             | - 0.2619  | - 0.0153 |
| Leverage                 |                          |           |           |                      |                         |                     | 1.0000   | 0.0820            | 0.1702             | - 0.1299  | 0.0102   |
| Capital Intensity        |                          |           |           |                      |                         |                     |          | 1.0000            | -0.3161            | 0.1187    | - 0.0899 |
| Age of Fixed Asset       |                          |           |           |                      |                         |                     |          |                   | 1.0000             | - 0.1789  | 0.1568   |
| Firm size                |                          |           |           |                      |                         |                     |          |                   |                    | 1.0000    | - 0.0095 |
| ROA                      |                          |           |           |                      |                         |                     |          |                   |                    |           | 1.000    |

**NA:** Not Applicable as our model does not associate between Tobin's Q, P/E Ratio and Corporate Governance Variables.

**Table 3b: Correlation Matrix (Iron & Steel Industry)**

|                          | Environmental Disclosure | Tobin's Q | P/E Ratio | Independent director | Ownership Concentration | Public Shareholding | Leverage | Capital Intensity | Age of Fixed Asset | Firm size | ROA     |
|--------------------------|--------------------------|-----------|-----------|----------------------|-------------------------|---------------------|----------|-------------------|--------------------|-----------|---------|
| Environmental Disclosure | 1.0000                   | -0.0087   | 0.0982    | 0.0042               | -0.0349                 | 0.0116              | -0.0958  | -0.0702           | 0.0915             | 0.5993    | 0.2779  |
| Tobin's Q                |                          | 1.0000    | 0.2334    | NA                   | NA                      | NA                  | -0.0402  | -0.4649           | -0.0680            | -0.1141   | 0.2541  |
| P/E Ratio                |                          |           | 1.0000    | NA                   | NA                      | NA                  | -0.1041  | 0.1064            | -0.0176            | 0.0912    | 0.0994  |
| Independent director     |                          |           |           | 1.0000               | -0.0398                 | 0.2112              | -0.1646  | 0.2937            | 0.1665             | -0.1638   | -0.2047 |
| Ownership Concentration  |                          |           |           |                      | 1.0000                  | -0.1255             | -0.1200  | -0.0900           | 0.2975             | 0.0409    | 0.0393  |
| Public Shareholding      |                          |           |           |                      |                         | 1.0000              | -0.1442  | 0.1378            | 0.1156             | -0.1495   | 0.0742  |
| Leverage                 |                          |           |           |                      |                         |                     | 1.0000   | 0.0059            | -0.2065            | 0.2357    | -0.1851 |
| Capital Intensity        |                          |           |           |                      |                         |                     |          | 1.0000            | 0.0831             | 0.1349    | -0.1374 |
| Age of Fixed Asset       |                          |           |           |                      |                         |                     |          |                   | 1.0000             | -0.2058   | 0.2797  |
| Firm size                |                          |           |           |                      |                         |                     |          |                   |                    | 1.0000    | -0.1023 |
| ROA                      |                          |           |           |                      |                         |                     |          |                   |                    |           | 1.0000  |

**NA:** Not Applicable as our model does not associate between Tobin's Q, P/E Ratio and Corporate Governance Variables.

**Table 3c: Correlation Matrix (IT&ES Industry)**

|                          | Environmental Disclosure | Tobin's Q | P/E Ratio | Independent director | Ownership Concentration | Public Shareholding | Leverage | Capital Intensity | Age of Fixed Asset | Firm size | ROA      |
|--------------------------|--------------------------|-----------|-----------|----------------------|-------------------------|---------------------|----------|-------------------|--------------------|-----------|----------|
| Environmental Disclosure | 1.0000                   | - 0.2420  | 0.2898    | 0.1287               | -0.0140                 | -0.2991             | - 0.1710 | -0.1068           | 0.0310             | 0.5904    | - 0.0956 |
| Tobin's Q                |                          | 1.0000    | 0.4115    | NA                   | NA                      | NA                  | 0.1527   | -0.5308           | 0.2084             | 0.0984    | 0.3392   |
| P/E Ratio                |                          |           | 1.0000    | NA                   | NA                      | NA                  | - 0.2056 | -0.3203           | 0.2247             | 0.0100    | 0.0170   |
| Independent director     |                          |           |           | 1.0000               | -0.1133                 | -0.0270             | - 0.1934 | 0.1530            | 0.0137             | 0.1232    | - 0.1355 |
| Ownership Concentration  |                          |           |           |                      | 1.0000                  | -0.4078             | - 0.0546 | -0.2887           | 0.2830             | -0.1353   | 0.1749   |
| Public Shareholding      |                          |           |           |                      |                         | 1.0000              | 0.0644   | 0.3023            | -0.0775            | -0.2305   | - 0.1728 |
| Leverage                 |                          |           |           |                      |                         |                     | 1.0000   | 0.0280            | 0.0060             | 0.1299    | 0.0171   |
| Capital Intensity        |                          |           |           |                      |                         |                     |          | 1.0000            | -0.4291            | 0.3874    | - 0.3133 |
| Age of Fixed Asset       |                          |           |           |                      |                         |                     |          |                   | 1.0000             | -0.3743   | 0.0245   |
| Firm size                |                          |           |           |                      |                         |                     |          |                   |                    | 1.0000    | - 0.3127 |
| ROA                      |                          |           |           |                      |                         |                     |          |                   |                    |           | 1.000    |

**NA:** Not Applicable as our model does not associate between Tobin's Q, P/E Ratio and Corporate Governance Variables.

**Table – 4: Result of Polluting Industry Fixed Effect Panel Data Model on the Impact of corporate Governance on Voluntary Environmental Disclosure**

| <b>Dependent Variable: Voluntary Environmental Disclosure</b> |                       |                        |                   |                |                                |                   |                |
|---------------------------------------------------------------|-----------------------|------------------------|-------------------|----------------|--------------------------------|-------------------|----------------|
| <b>Explanatory Variables</b>                                  | <b>Predicted sign</b> | <b>Cement Industry</b> |                   |                | <b>Iron and Steel Industry</b> |                   |                |
|                                                               |                       | <i>Coefficient</i>     | <i>Std. Error</i> | <i>t ratio</i> | <i>Coefficient</i>             | <i>Std. Error</i> | <i>t ratio</i> |
| <b>Constant</b>                                               |                       | -1.18825               | 10.4885           | -0.1133        | 26.7220                        | 12.3330           | 2.167***       |
| <b>D<sub>1</sub></b>                                          | +                     | 1.85836                | 1.01892           | 1.824*         | 1.76342                        | 0.446989          | 3.945***       |
| <b>D<sub>2</sub></b>                                          | +                     | 3.68810                | 1.42085           | 2.596**        | 3.85373                        | 1.11948           | 3.442***       |
| <b>D<sub>3</sub></b>                                          | +                     | 4.53872                | 1.52369           | 2.979***       | 5.44275                        | 1.34171           | 4.057***       |
| <b>D<sub>4</sub></b>                                          | +                     | 6.06164                | 1.70869           | 3.548***       | 7.45235                        | 1.33939           | 5.564***       |
| <b>Independent Director</b>                                   | +                     | 0.0563295              | 0.0882322         | 0.6384         | 0.0168099                      | 0.0378341         | 0.4443         |
| <b>Ownership Concentration</b>                                | –                     | - 0.073909             | 0.111091          | -0.6653        | - 0.0777327                    | 0.0358796         | -2.166**       |
| <b>CEO Duality</b>                                            | –                     | 0.936135               | 1.64541           | 0.5689         | 0.973530                       | 1.36898           | 0.7111         |
| <b>Public Shareholding</b>                                    | +                     | 0.133616               | 0.0801738         | 1.667*         | 0.0702927                      | 0.0913340         | 0.7696         |
| <b>Environmental Committee</b>                                | +                     | 4.68181                | 2.43317           | 1.924*         | 0.287120                       | 0.968645          | 0.2964         |
| <b>Leverage</b>                                               | –                     | -0.298482              | 0.127648          | -2.338**       | -1.16776                       | 0.915226          | -1.276         |
| <b>Capital Intensity</b>                                      | +/-                   | 0.0484918              | 0.0352307         | 1.376          | -0.0156609                     | 0.0165913         | 0.9439         |
| <b>Age of Fixed Assets</b>                                    | +/-                   | 14.1262                | 4.47957           | 3.153***       | 1.74624                        | 8.13718           | 0.2146         |
| <b>Firm Size</b>                                              | +                     | 1.30579                | 1.11753           | 1.168          | 0.234250                       | 1.39993           | 0.1673         |
| <b>ROA</b>                                                    | +                     | -0.0016                | 0.0009            | -1.783*        | -0.0430030                     | 0.0239064         | -1.799*        |
| <b>R-squared</b>                                              |                       | 0.91                   |                   |                | 0.98                           |                   |                |

|                                                                                                                                                                                                                                               |  |                                                                                                    |                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Adjusted R-squared</b>                                                                                                                                                                                                                     |  | 0.87                                                                                               | 0.97                                                                                               |
| <b>F Value</b>                                                                                                                                                                                                                                |  | 21.79125***                                                                                        | 120.1453***                                                                                        |
| <b>Durbin-Watson</b>                                                                                                                                                                                                                          |  | 2.051249                                                                                           | 1.979590                                                                                           |
| <b>N (observations)</b>                                                                                                                                                                                                                       |  | 26                                                                                                 | 29                                                                                                 |
| <b>Test for Differing Group Intercepts</b>                                                                                                                                                                                                    |  | Null hypothesis: The groups have a common intercept<br>Test statistic: $F(25, 86) = 7.56281^{***}$ | Null hypothesis: The groups have a common intercept<br>Test statistic: $F(25, 86) = 57.4657^{***}$ |
| <p><b>The calculation procedure in GRETl follows Robust (Heteroscedasticity Autocorrelation Consistent) standard errors.</b></p> <p><b>*: <math>p &lt; 0.10</math>;    **: <math>p &lt; 0.05</math>;    ***: <math>p &lt; 0.01</math></b></p> |  |                                                                                                    |                                                                                                    |

**Table – 5: Result of Non-polluting Sector and Pooled Sample of Polluting Sector Fixed Effect Panel Data Model on the Impact of corporate Governance on Voluntary Environmental Disclosure**

| <b>Dependent Variable: Voluntary Environmental Disclosure</b> |                       |                           |                   |                |                                                  |                   |                |
|---------------------------------------------------------------|-----------------------|---------------------------|-------------------|----------------|--------------------------------------------------|-------------------|----------------|
| <b>Explanatory Variables</b>                                  | <b>Predicted sign</b> | <b>IT&amp;ES Industry</b> |                   |                | <b>Pooled Sample (Cement + Iron &amp; Steel)</b> |                   |                |
|                                                               |                       | <i>Coefficient</i>        | <i>Std. Error</i> | <i>t ratio</i> | <i>Coefficient</i>                               | <i>Std. Error</i> | <i>t ratio</i> |
| <b>Constant</b>                                               |                       | 29.2525                   | 9.08570           | 3.220***       | 3.05446                                          | 4.90352           | 0.6229         |
| <b>D<sub>1</sub></b>                                          | +                     | -0.190315                 | 1.18449           | -0.1607        | 1.57438                                          | 0.707149          | 2.226**        |
| <b>D<sub>2</sub></b>                                          | +                     | 2.78364                   | 1.32566           | 2.100**        | 3.45980                                          | 0.744354          | 4.648***       |
| <b>D<sub>3</sub></b>                                          | +                     | 4.41456                   | 1.47371           | 2.996***       | 4.66336                                          | 0.798448          | 5.841***       |
| <b>D<sub>4</sub></b>                                          | +                     | 5.22371                   | 1.47289           | 3.547***       | 6.22755                                          | 0.864765          | 7.201***       |
| <b>Independent Director</b>                                   | +                     | 0.0505637                 | 0.0626938         | 0.8065         | 0.03306                                          | 0.034028          | 0.9717         |
| <b>Ownership Concentration</b>                                | –                     | -0.114301                 | 0.0548441         | -2.084**       | -0.06183                                         | 0.042648          | -1.450*        |

[illegible]



**Table – 6a: Result of Impact of Voluntary Environmental Disclosure on Market Based Measures in Cement Industry**

| <b>Sample: Cement Industry</b>             |                       |                                                     |                   |                |                                                     |                   |                |
|--------------------------------------------|-----------------------|-----------------------------------------------------|-------------------|----------------|-----------------------------------------------------|-------------------|----------------|
| <b>Explanatory Variables</b>               | <b>Predicted sign</b> | <b>Dependent Variable: Tobin's Q</b>                |                   |                | <b>Dependent Variable: P/E Ratio</b>                |                   |                |
|                                            |                       | <i>Coefficient</i>                                  | <i>Std. Error</i> | <i>t ratio</i> | <i>Coefficient</i>                                  | <i>Std. Error</i> | <i>t ratio</i> |
| <b>Constant</b>                            |                       | 11.6142                                             | 2.07891           | 5.587***       | 65.4143                                             | 25.3726           | 2.5782**       |
| <b>D<sub>1</sub></b>                       | +                     | 1.7087                                              | 0.496704          | 3.440***       | 42.098                                              | 6.37421           | 6.6044***      |
| <b>D<sub>2</sub></b>                       | +                     | 1.17567                                             | 0.465627          | 2.5249**       | 23.3081                                             | 10.7966           | 2.1588**       |
| <b>D<sub>3</sub></b>                       | +                     | 1.15162                                             | 0.477799          | 2.4102**       | 2.29225                                             | 7.93906           | 0.2887         |
| <b>D<sub>4</sub></b>                       | +                     | 1.36773                                             | 0.533926          | 2.5616**       | 2.19634                                             | 7.87275           | 0.2790         |
| <b>Environmental Disclosure</b>            | +                     | -0.0109138                                          | 0.0306557         | -0.3560        | 0.552816                                            | 0.480692          | 1.1500*        |
| <b>Leverage</b>                            | +                     | 0.162029                                            | 0.0630978         | 2.5679**       | -0.852909                                           | 0.541489          | -1.5751*       |
| <b>Capital Intensity</b>                   | +/-                   | -0.0148012                                          | 0.007520          | -1.9682*       | 0.0883265                                           | 0.121172          | 0.7289         |
| <b>Age of Fixed Assets</b>                 | +/-                   | -1.28584                                            | 1.74336           | -0.7376        | -45.8081                                            | 19.6488           | -2.3313**      |
| <b>Firm Size</b>                           | +                     | -1.48611                                            | 0.335742          | -4.42***       | -8.90465                                            | 4.0628            | -2.1918**      |
| <b>ROA</b>                                 | +                     | 0.00032                                             | 0.000565          | 0.5664         | 0.0221381                                           | 0.004558          | 4.8570***      |
| <b>R-squared</b>                           |                       | 0.782966                                            |                   |                | 0.515663                                            |                   |                |
| <b>Adjusted R-squared</b>                  |                       | 0.698564                                            |                   |                | 0.327310                                            |                   |                |
| <b>F Value</b>                             |                       | 9.276634***                                         |                   |                | 2.73775***                                          |                   |                |
| <b>Durbin-Watson</b>                       |                       | 1.862743                                            |                   |                | 2.33902                                             |                   |                |
| <b>N (observations)</b>                    |                       | 26                                                  |                   |                | 26                                                  |                   |                |
| <b>Test for Differing Group Intercepts</b> |                       | Null hypothesis: The groups have a common intercept |                   |                | Null hypothesis: The groups have a common intercept |                   |                |

|                                                                                                                                                                                                                                               |  |                                            |                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------------------------|------------------------------------------|
|                                                                                                                                                                                                                                               |  | Test statistic: $F(25, 90) = 4.5498^{***}$ | Test statistic: : $F(25, 90) = 0.961577$ |
| <p><b>The calculation procedure in GRETl follows Robust (Heteroscedasticity Autocorrelation Consistent) standard errors.</b></p> <p><b>*: <math>p &lt; 0.10</math>;    **: <math>p &lt; 0.05</math>;    ***: <math>p &lt; 0.01</math></b></p> |  |                                            |                                          |

**Table – 6b: Result of Impact of Voluntary Environmental Disclosure on Market Based Measures in Iron & Steel Industry**

| <b>Sample: Iron and Steel Industry</b> |                       |                                      |                   |                |                                      |                   |                |
|----------------------------------------|-----------------------|--------------------------------------|-------------------|----------------|--------------------------------------|-------------------|----------------|
| <b>Explanatory Variables</b>           | <b>Predicted sign</b> | <b>Dependent Variable: Tobin's Q</b> |                   |                | <b>Dependent Variable: P/E Ratio</b> |                   |                |
|                                        |                       | <i>Coefficient</i>                   | <i>Std. Error</i> | <i>t ratio</i> | <i>Coefficient</i>                   | <i>Std. Error</i> | <i>t ratio</i> |
| <b>Constant</b>                        |                       | 20.1856                              | 6.26089           | 3.224***       | -79.8021                             | 53.9787           | -1.4784        |
| <b>D<sub>1</sub></b>                   | +                     | 2.32785                              | 0.530641          | 4.387***       | 2.8983                               | 3.67225           | 0.7892         |
| <b>D<sub>2</sub></b>                   | +                     | 1.73154                              | 0.417861          | 4.144***       | -6.94414                             | 6.93763           | -1.0009        |
| <b>D<sub>3</sub></b>                   | +                     | 1.4038                               | 0.486101          | 2.888***       | -11.901                              | 9.28621           | -1.2816        |
| <b>D<sub>4</sub></b>                   | +                     | 1.33729                              | 0.604503          | 2.2122**       | -12.6801                             | 9.48158           | -1.3373        |
| <b>Environmental Disclosure</b>        | +                     | -0.0408394                           | 0.0646361         | -0.6318        | 1.68854                              | 1.01569           | 1.6625*        |
| <b>Leverage</b>                        | +                     | -0.22317                             | 0.33002           | -0.6762        | -0.798544                            | 5.98503           | -0.1334        |
| <b>Capital Intensity</b>               | +/-                   | -0.0147799                           | 0.0102597         | -1.4406        | 0.12804                              | 0.150412          | 0.8513         |
| <b>Age of Fixed Assets</b>             | +/-                   | -5.06694                             | 3.41716           | -1.4828        | 37.1212                              | 42.1702           | 0.8803         |
| <b>Firm Size</b>                       | +                     | -2.19704                             | 0.875189          | -2.510**       | 5.73144                              | 5.4868            | 1.0446**       |
| <b>ROA</b>                             | +                     | 0.002940                             | 0.00959           | 0.3065         | 0.0710964                            | 0.09088           | 0.7823         |
| <b>R-squared</b>                       |                       | 0.771065                             |                   |                | 0.475872                             |                   |                |
| <b>Adjusted R-squared</b>              |                       | 0.688994                             |                   |                | 0.287977                             |                   |                |

|                                                                                                                                                                                                                              |  |                                                                                                    |                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>F Value</b>                                                                                                                                                                                                               |  | 9.39509***                                                                                         | 2.532652***                                                                                         |
| <b>Durbin-Watson</b>                                                                                                                                                                                                         |  | 1.799349                                                                                           | 1.922929                                                                                            |
| <b>N (observations)</b>                                                                                                                                                                                                      |  | 29                                                                                                 | 29                                                                                                  |
| <b>Test for Differing Group Intercepts</b>                                                                                                                                                                                   |  | Null hypothesis: The groups have a common intercept<br>Test statistic: $F(28, 106) = 7.6753^{***}$ | Null hypothesis: The groups have a common intercept<br>Test statistic: $F(28, 106) = 2.79696^{***}$ |
| <b>The calculation procedure in GRETL follows Robust (Heteroscedasticity Autocorrelation Consistent) standard errors.</b><br><i>*: <math>p &lt; 0.10</math>; **: <math>p &lt; 0.05</math>; ***: <math>p &lt; 0.01</math></i> |  |                                                                                                    |                                                                                                     |

**Table – 6c: Result of Impact of Voluntary Environmental Disclosure on Market Based Measures in IT & ES Industry**

| <b>Sample: IT&amp;ES Industry</b> |                       |                                      |                   |                |                                      |                   |                |
|-----------------------------------|-----------------------|--------------------------------------|-------------------|----------------|--------------------------------------|-------------------|----------------|
| <b>Explanatory Variables</b>      | <b>Predicted sign</b> | <b>Dependent Variable: Tobin's Q</b> |                   |                | <b>Dependent Variable: P/E Ratio</b> |                   |                |
|                                   |                       | <i>Coefficient</i>                   | <i>Std. Error</i> | <i>t ratio</i> | <i>Coefficient</i>                   | <i>Std. Error</i> | <i>t ratio</i> |
| <b>Constant</b>                   |                       | 44.7659                              | 13.9615           | 3.206***       | 30.6728                              | 7.75736           | 3.9540***      |
| <b>D<sub>1</sub></b>              | +                     | 11.4218                              | 3.83125           | 2.981***       | 8.38282                              | 1.85219           | 4.5259***      |
| <b>D<sub>2</sub></b>              | +                     | 7.89232                              | 3.1951            | 2.4701**       | 10.0274                              | 2.49381           | 4.0209***      |
| <b>D<sub>3</sub></b>              | +                     | 9.33717                              | 3.0668            | 3.045***       | 7.27033                              | 2.42625           | 2.9965***      |
| <b>D<sub>4</sub></b>              | +                     | 10.8414                              | 3.68431           | 2.943***       | 6.6503                               | 2.27192           | 2.9272***      |
| <b>Environmental Disclosure</b>   | +                     | -0.484433                            | 0.226855          | -2.135**       | 0.24517                              | 0.12583           | -1.9483*       |
| <b>Leverage</b>                   | +                     | 0.815087                             | 2.6655            | 0.3058         | -2.99291                             | 2.04726           | -1.4619        |
| <b>Capital Intensity</b>          | +/-                   | -0.497718                            | 0.217171          | -2.292**       | -0.08413                             | 0.06599           | -1.2749        |

[illegible]

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## Annexure - 1

### Environmental Disclosure Grid

| Sl No.    | Items                                                                                        | Scale of Scoring            | Map to GRI |
|-----------|----------------------------------------------------------------------------------------------|-----------------------------|------------|
| <b>A)</b> | <b>Strategy and Analysis (Maximum Score 5)<br/>Scale A</b>                                   | <b>*Follow Rating</b>       |            |
| 1         | Statement of senior decision maker on environmental policy                                   | 0-1                         | GN 1.1     |
| 2         | A statement on formal management system regarding key environmental risk and impacts.        | 0-1                         | GN 1.2     |
| 3         | Mention about key events, achievements, awards and failures during reporting period.         | 0-1                         | GN 1.1     |
| 4         | A statement on organization's main environmental challenges and goals in coming 3 – 5 years. | 0-1                         | GN 1.1     |
| 5         | Involvement to environmental organization/association to improve environmental practices     | 0-1                         | GN 4.13    |
| <b>B)</b> | <b>Assurance(Max score 1)<br/>Scale A</b>                                                    | <b>*Follow Rating</b>       |            |
| 1         | Independent verification or assurance on company's environmental performance                 | 0-1                         | GN 3.13    |
| <b>C)</b> | <b>Governance, commitments and engagements (Max score 3)<br/>A</b>                           | <b>*Follow Rating Scale</b> |            |
| 1         | Existence of an environmental committee in the board                                         | 0-1                         | GN 4.1     |
| 2         | Periodic verifications or audits on environmental performance and/or systems                 | 0-1                         | GN 4.9     |
| 3         | Adoption of externally developed environmental charters or principles or other initiatives   | 0-1                         | GN 4.12    |
| <b>D)</b> | <b>Laws and Regulations Conformity (Max score 6)<br/>Scale B</b>                             | <b>**Follow Rating</b>      |            |
| 1         | Litigations, actual and potential fines                                                      | 0-3                         | EN 28      |
| 2         | Order to conform corrective actions                                                          | 0-3                         |            |
| <b>E)</b> | <b>Expenditures and Risk (Max score 12)<br/>Scale B</b>                                      | <b>**Follow Rating</b>      |            |

|                                                                   |                                                                                                                               |                        |                   |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------|
| 1                                                                 | Investment on technologies, researches and/or innovations to enhance environmental performance and efficiency                 | 0-3                    | EN 30             |
| 2                                                                 | Environmental operating costs                                                                                                 | 0-3                    |                   |
| 3                                                                 | A statement on governance mechanism in place to specifically identify and manage environmental risks and opportunities.       | 0-3                    | GN 1.2            |
| 4                                                                 | Provisions for future expenditures                                                                                            | 0-3                    |                   |
| <b>F) Environmental Performance Indicators (Maximum Score 33)</b> |                                                                                                                               | <b>**Follow Rating</b> |                   |
| <b>Scale B</b>                                                    |                                                                                                                               |                        |                   |
| 1                                                                 | Direct and indirect energy consumption by primary energy source                                                               | 0-3                    | EN 3,4            |
| 2                                                                 | Materials used and percentage of recycled input materials                                                                     | 0-3                    | EN 1,2            |
| 3                                                                 | Energy saved due to conservation, efficiency improvement and initiatives towards renewable energy based products and services | 0-3                    | EN 5,6            |
| 4                                                                 | Water usage and water usage efficiency                                                                                        | 0-3                    | EN 8,9,10         |
| 5                                                                 | Total direct and indirect greenhouse gas emissions by weight                                                                  | 0-3                    | EN 16,17          |
| 6                                                                 | Initiatives to reduce greenhouse gas emissions and reduction achieved                                                         | 0-3                    | EN 18             |
| 7                                                                 | NOx, SOx and other significant air emissions and ozone depleting substances by type and weight                                | 0-3                    | EN 19,20          |
| 8                                                                 | Volume of significant land, water and air discharges, releases and spills                                                     | 0-3                    | EN 21,23          |
| 9                                                                 | Waste generation and its disposal and treatment                                                                               | 0-3                    | EN 22, 24, 25     |
| 10                                                                | Initiatives to reduce/mitigate environmental impacts of products and services                                                 | 0-3                    | EN 26,27          |
| 11                                                                | Natural resources conservations                                                                                               | 0-3                    | EN 11,12,13,14,15 |
| <b>Total Maximum Possible Score is 60</b>                         |                                                                                                                               |                        |                   |

**\*Rating Scale A**

- 1: Yes, the item is mentioned
- 0: No, the item is not mentioned

**\*\*Rating Scale B**

- 3: Item described in monetary or quantitative terms
- 2: Item described specifically
- 1: Item discussed in general

## Annexure – 2

### Boxplots of Voluntary Environmental Disclosure Scores

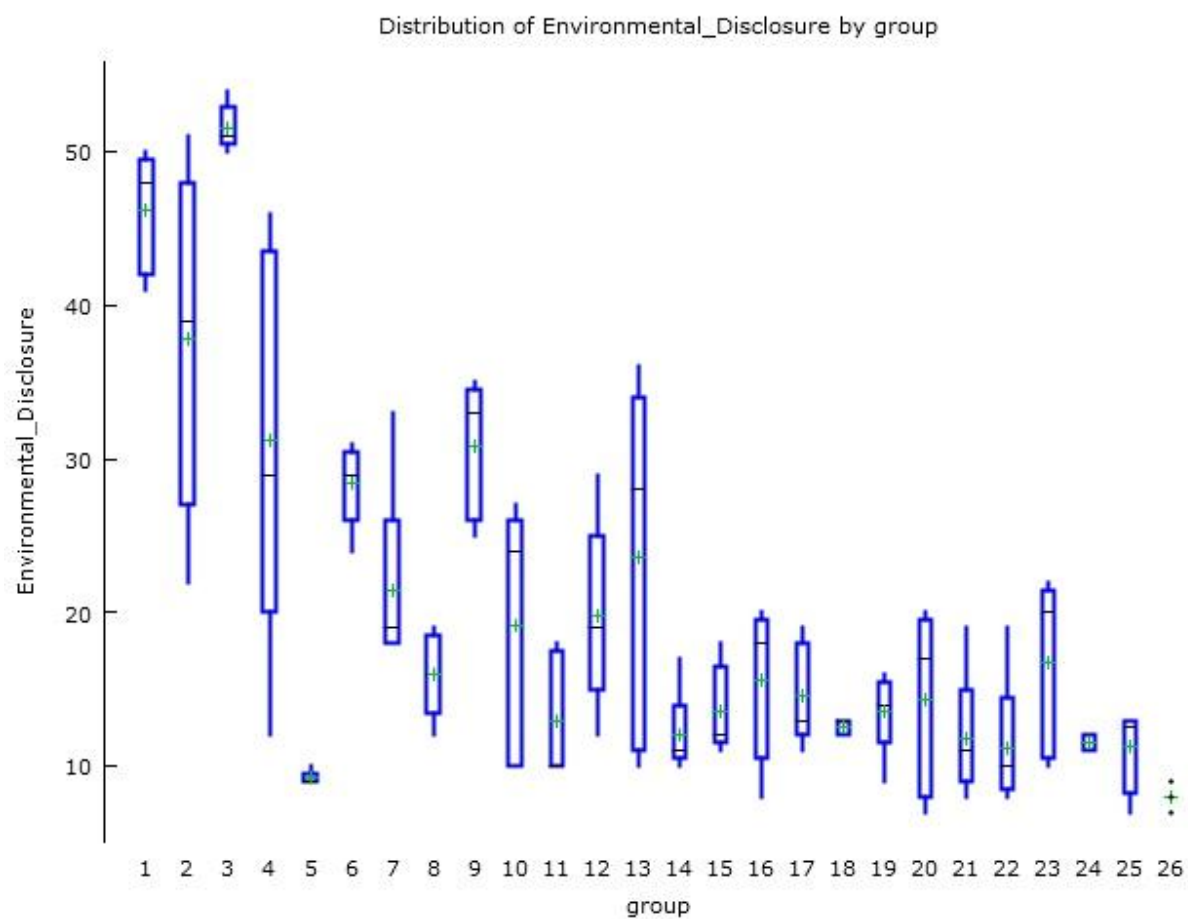
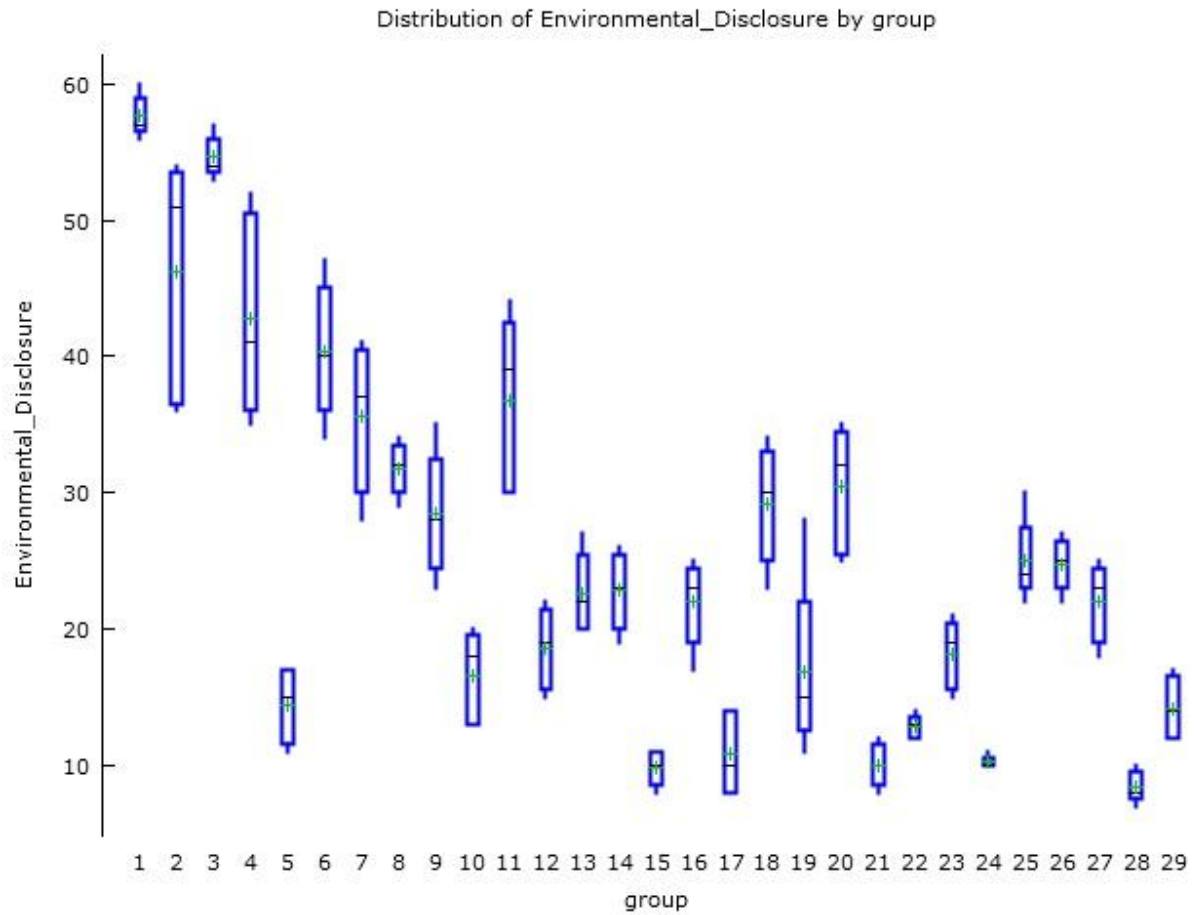
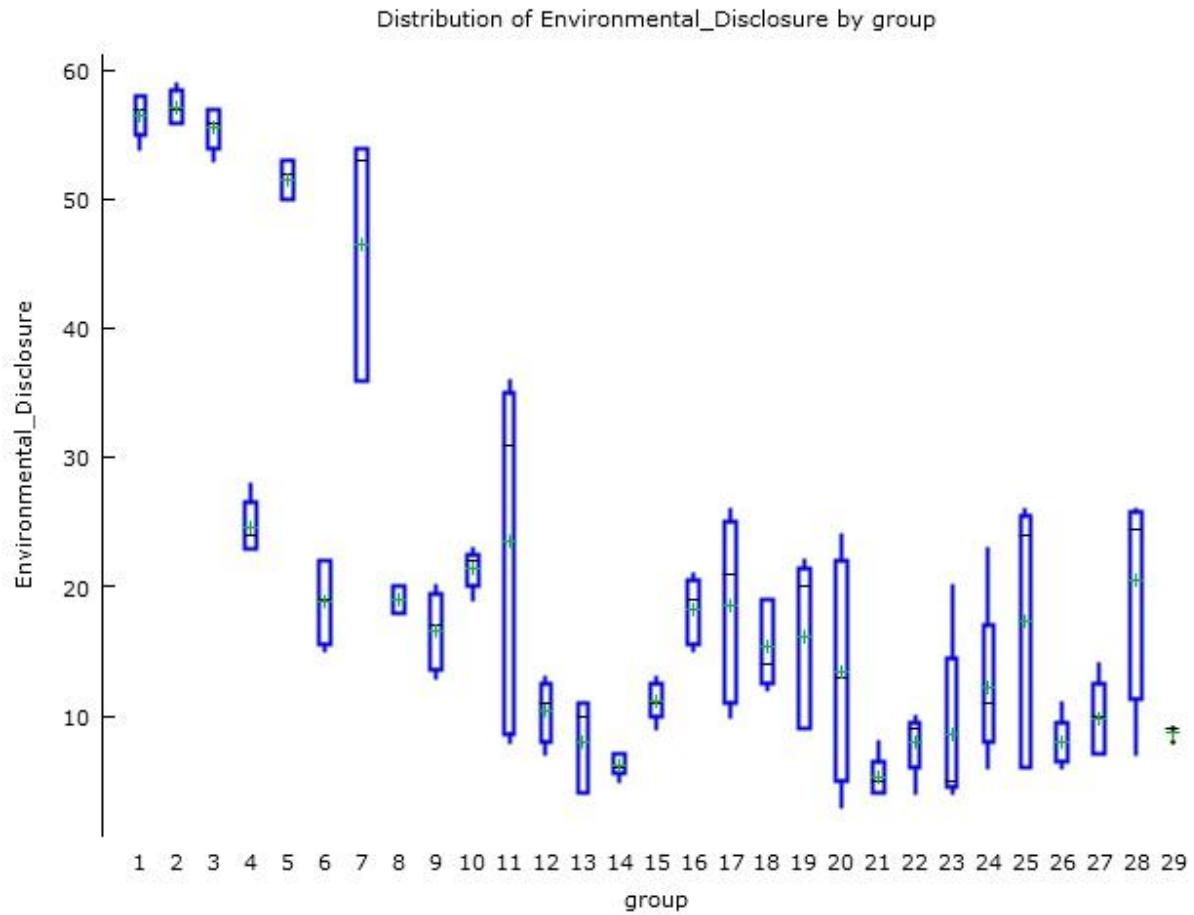


Figure 1: Boxplots for environmental disclosure score of cement industry sample



**Figure 2: Boxplots for environmental disclosure score of iron and steel industry sample**



**Figure 3: Boxplots for environmental disclosure score of ITES industry sample**